

Thesis in

*Information Science, City Sciences, Computer Science, Electrical and Computer Engineering,
Systems and Software Engineering, Socio-Technical Systems, High-Performance Networking,
Environmental Engineering, with Artificial Intelligence and Human-Computer Interaction*

‘Glimpse-Versioned’

“The Necessity, Feasibility, Intelligence, Interaction, and Actualization of the Applications in Environmental Computation”

** The thesis is mainly relevant to Computer Science and partially related to
other Applied Sciences and Engineering Fields **

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NOTE:

This is a version of Beta 0.9, so the paper's outline in thesis development still exists.

ABSTRACT

1.01 - Purpose of Writing

The purpose of writing this paper on the technology of Environmental Computation (EC) is to introduce its fundamental notion and then fancy the applicable scenarios to approach the possibility of actualizing the tangible entities – the artifacts of the EC's embedded software systems and the computational nodes among the environments, to eventually verify the prerequisites and necessity and validate the intelligence and feasibility of the development prospect of the Environmental Computation's technology. By brainstorming, innovating, conceiving, critical thinking, discussing, and comprehending through the entire thesis development, ultimately, this paper shall possess a bit of referencing value and instructive meaningfulness to be a beacon to materialize the feasible applications in the tomorrow of Environmental Computation whatever in academia and industries related.

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INTRODUCTION

2.01 - Background

Before introducing the term technology of Environmental Computation, we must first discuss the technological terms of Smart Cities/Homes. Actually, in essence, these two kinds of technologies, Environmental Computation and Smart Cities/Homes, do not radically have many embodying discrepancies, possessing featured similarities in principles or rationales of usability and engineering as Human-Centered applications, which are like the geographically deployed facilities/devices, containing the embedded software systems, providing the way in Human-Computer Interactions, having a main or multiple end users' dashboards/interfaces, and designed for particular task-achievement and problem-solving functionalities, which satisfy the people's needs in their daily lives today. Nevertheless, differentiations exist between the two tech terms with the relevant applied reasoning and mattering, one cannot simply substitute another because of their distinguish pros and cons in respective reality from the applicable scenes. Well, let's initially see what exactly the scientific definition is about the terms of Smart Cities/Homes for beneficially introducing the term Environmental Computation.

The term "Smart City" refers to the use of technology and data to improve the quality of life

for its residents, enhance sustainability, and optimize the efficiency of urban services. It involves the integration of various digital technologies, such as the Internet of Things (IoT), big data analytics, and artificial intelligence, to manage and automate different aspects of urban life, including transportation, energy, waste management, security, and more.

On the other hand, a "Smart Home" refers to a living space equipped with various connected devices and systems that can be controlled remotely or automated to enhance comfort, convenience, and energy efficiency. Smart Home technology often includes smart appliances, thermostats, security systems, lighting controls, and more, which can be managed through a central hub or mobile app.

However, the above needs to be explained dedicatedly, with more details needed in paradigmatic or restricted application scenarios to point out the technology's feasible and applied limitations and constraints. We shall at least fathom that a systemic structured entity of Smart Cities/Homes consisting of whatever the artifact of software or aspect of hardware as a Human-Centered usable device or facility destined to serve a goal chiefly, which means each of Smart Cities/Homes things exclusively fulfill the unique tasks derived from the requirements of the people as one or group end users. For instance, a ticket kiosk in self-service mode, airport check-in, or tourism informational site must intently achieve the task objective, of ticketing the exact stuff to the end user. This leads to a situation in which a Smart Cities/Homes unit would be mandatory to handle solely one task or a series of functions that meet the requirements of the end-users. The limitations and constraints inevitably affiliate with the unit itself in embodying and engineering the actualizations, that's why nowadays we see great bunches of diverse units, nodes, devices, or facilities as the Smart Cities/Homes entities running embedded software systems. Metaphorically, as warriors, every one of them fights their own battles alone even miserably as a non-legion, there's no authentically real-time, deep, and interactive collaboration between one and another of the Smart Cities/Homes units, even though, from the theoretical observational angle, they may share, bargain, and communicate the data each other on the level of networking streams or sessions in general.

Reasonably, according to the opinion that the entities as devices or facilities of Smart Cities/Homes do interlace each other in data exchange and processing, we are willing to admit that they work together in some unifying. But, in most situations of reality, they often do not synchronously interact and collaborate even though the manufacturers, experts, or scientists manifested how Smart Cities/Homes will benefit people's daily lives and bring a brand-new era of "Everything" in "Smart" enough to work together, forging a better tomorrow. However, the technological development reflected by the applicable circumstances nowadays in tech Smart Cities/Homes did not perform as expected, though we cannot assert it's currently in a debacle. Here we can generally see the thorough development or evolvement data statistically.

> Insert the data presentation here:

Based on the above statistical reality, we realized that many difficulties and obstacles exist during the development of Smart Cities/Homes technology. The reasons remain complicated;

probably, the lack of investment in monetary or labor empowerment, possibly, the incapacity in knowledge or craftiness of the technicians and practitioners in software development and hardware integration, and maybe, the immature prerequisite of the environments that deploy the devices or facilities of Smart Cities/Homes in civil or electrical engineering. Besides, more significantly, there are many non-ignorable irrelevances in the usability and experience of each Smart Cities/Homes device or facility entity. For example, two sets of Smart Home facilities, produced by discrepant manufacturers, that are deployed in respective apartments, possibly never even interchange their on-processing or archived data, not to mention the interoperability between them. For another instance, in the same airport, a self-service check-in kiosk doesn't collaborate with another kiosk simply because they belong to different airlines. Furthermore, in an extreme applicational scene, an auto-ticketing and auto-check-in terminal with highly advanced Smart City tech features owned by the same airways company may still lack communications in not only data coordination and cross-usability, which reasons depend on various factors such as embedded software apps or operating systems incompatibility and local/wide networking patterns in the initial architecture design and hardware computation capacities.

Although the involved industries, manufacturers, researchers, and scientists confidently announce that the technology Smart Cities/Homes with the applications shall closely work together at best or at least be capable of teaming up to target the mutual tasking to deliver the fulfillment to the end users in applicational scenarios, the cases before described already profoundly proved there's still a gap between the reality and purposing. In development, countless issues, cruxes, and problems in tech Smart Cities/Homes require solving and breakthroughs, particularly in the applicational implementation. However, we ought to realize that a way of resolving the materialized objects one by one of the Smart Cities/Homes entities in whatever their systems architecture design and human-computer interactions may not be the ultimate approach. So, here's a raised question: Is there a series of solutions, mechanisms, and designs, derived from reliable methodology, to wisely walk out of the predicament that draws back the applicational Smart Cities/Homes research and optimization in sustainable development?

We believe the answer is affirmative and exists in our mindset which needs to wisely figure it out via critical thinking in the systems architecture design, user factors research, user experience prototyping, embedded applications modeling, software engineering, networks patterns scheming, collaborative mechanisms, and so on. Yet, this above pinpointing may just roughly unveil some stuff that shall still be kind of superficial from a glimpse, which means the difficulties, in the evolution of the applications in Smart Cities/Homes and the technology, is a complex and gradual process, depending on a great amount of theoretical research and empirical implementation with indispensable experiments. Nonetheless, the writing purpose of the thesis isn't seeking a way for the optimization of the technological theories and applicable actualization in Smart Cities/Homes, which is to introduce a brand new but tenably wiser solution that fulfills the same objective of the technology of Smart Cities/Homes in applications - Environmental Computation. So, with that, let's head to the motivation of the thesis to see everything unambiguously.

2.02 - Motivation

We apprehend that solely counting on this paper may just find an achievable way to build an all-in-one solution to identify and overcome the cruxes and optimize the progress of the technology in Smart Cities/Homes. Nonetheless, in our brief, the thesis may discover the essentiality that matters in the effectiveness and actualization of Environmental Computation with successive research and experiments. We think, more or less, by initially fathoming on the presently relative scenarios such as the applications in Smart Cities/Homes, profoundly, the terminology of Environmental Computation (EC) and methodology of the actualization in EC applications can be scoped, identified, defined, specified, and concretized, which can last power the evolution of the contemporary sociotechnical systems facilitating the people's needs derived from their daily business or/and recreational activities.

By authoring this paper, the motivation for our research on the topic of Environmental Computation is to erect a set of rationalities that convincingly corroborates the necessity of existence, the prerequisites of development, the feasibility of the applicational actualization, and the convergent possibility with the artificial intelligence-driven mechanism in the complex high-performance computing sociotechnical systems, fixing a basement as a tenable outset to move on the following intensive experiments in simulating and prototyping the various tangibly applicational artifacts or entities under the circumstances of different applied scenarios in Environmental Computation (EC) such as the above exemplified the restaurant industry using EC to intrigue the dining willingness from the potential consumers in a specific surrounding atmosphere at the proper moment in people's daily lives. Estimated, with accumulatively tentative EC applications prototyped, in the end, a series of repeatedly experimented methodologies composing the standardizable approaches and techniques will be naturally but also scientifically formed to benefit the after-research and foresee-actualization in Environmental Computation's technical and commercial development.

Purposely, as we reach the rapid evolution of Environmental Computation (EC) technology and the situation that ubiquitous EC entities exist in people's lives around the world, there were quite a few rigorously validated and verified methods, mechanisms, techniques, and theories shall be spontaneously formed via the scientists and practitioners' researching toils. Under the best circumstances, in methods, amount of libraries and frameworks in programming for the embedded software/systems running in the EC entities should be increasingly developed; in mechanisms, a variety of specific characterized operating systems with the applicable Artificial Intelligence augments for the EC units ought to be scoped and settled; in techniques, a bunch of utilities or extensions championing the software/hardware of the EC applicational stuff shall be innovated and set; and in theories, a series of concepts, notions, rationales, and methodologies would be standardized and defined eventually. At that time, we may say the technology of Environmental Computation in its theoretical and applicational actualization has been successful. Also, our motivation for researching through this thesis development has been superbly functional. Very

simply speaking, we sincerely wish our attempt to author the paper with the research motivation can be expectantly useful to solidify a good and feasible enough genesis in the upcoming revolutionary development of the technology in Environmental Computation, hoping, at least, this writing can be one of the works that benefit the research on the relevant topic in Environmental Computation that other participants contributing the same efforts in academia and industries.

2.03 - Understandability and Classification

For “Environmental Computation”, respectively in literal understanding, the phrase “Computation” means there must be a task-oriented and data-driven computational system integrating a cluster of designed algorithms and specified mechanisms to realistically contrive an expected outcome that ultimately fulfills the specific applicational requirements of scoped Human-Centered living and industrial demands via modern Human-Computer Interaction tactics. While the phrase “Environmental” means the “Computation” shall be rationally constrained by the feature of the very “Environment”, which generates a “featured” capability of assisting human beings’ living needs and productive purposes. As unified, the terminology “Environmental Computation” implies it’s a kind of “Environment-Feature-Driven Computational Methodology” that rigorously obeys the pivotal demands elicited from the people who are immersively experiencing and purposely working in the very environment armed with the power of present-day technologies in computational competences.

Every environment owns one pivotal “feature”, if we jump out of the observational angles in the complexity to simply regard one environment as an atomic entity, we can easily identify that each one of the environments always carries its core-defined and designed “functionality”, which is essentially an “applicable characteristic” and also a services-objective mechanism aiming to human society living and productive needs. We do say that’s why people go to specific places for specific requirements fulfillment; e.g. A Costco supermarket is built for the people who like to purchase their living supplements, a campus parking lot is constructed for the students and faculty or staff to place their vehicles, and a manufacturing dock is erected for the workers to facilitate their logistic routine. These vivid instances demonstrate the easily understandable concept of the importance of clearly pinpointing and fathoming what exactly the “Feature” of an “Environment” is.

Nevertheless, the peculiar “features” of the respective environments in detailed delimitation on technologically applicable topics and concentrations remain somewhat complicated. For instance, realistically, a complex building might have quite a few dissimilar “features” based on discrepant applicational scenarios derived from the requirements of respectively classified people such as the staff of the companies leasing the offices in the complex, the maintenance personnel of the property like the security or janitors, or the visitors. Seemingly, according to this illustrated analysis, the scope and definition of a “feature” of the very environment became paradoxical to raise a question: Shall an environment legitimately own multiple “features”? The answer is “No”. Always, an environment does have an “exclusively pivotal” feature, the notion is pretty much like we work on the stage of

Requirement Engineering in SDLC (Software Development Life Cycle), there's a pair of analyses, the "Functional Requirements" and "Non-Functional Requirements". For example, a Pet-Rescue Charity Website or App ...

2.04 - Rationality and Scenario Glimpses

The rationality in Environmental Computation is not so hard to scope and specify, which is explicitly a sort of technology concentrating on various applications with the computing capacity running in specific environments under discrepant circumstances integrating relevant requirements intrigued by human cognition in social activities. Further from a microscope perspective, there needs to be a system following the Human-Centered technology principles and applying the Human-Computer Interaction availabilities to act the intended fulfillment satisfying the end user's requested demands as a Socio-Technical entity via a computing mechanism based on specified applicable scenarios actualized in scoped locations. Nonetheless, not to mention the complexity of Computation in both theory and reality in Computer Science and Applicational Experiments, the environmental factors remain highly complicated in the definition that would encounter limitations in present sciences and technologies, restrictions in existing social cultures and policies, and uncertainty in constantly changing lifestyles and business modes of human beings. Thus, to say nowadays we are already capable of precisely defining the notion and rationale of Environmental Computation is too subjective and even arbitrary, at least remaining unrealistic in general. Yet, narrowing down the viewpoint of specifically applicational research in Environmental Computation's actualizations is superbly feasible and also a valuable necessity. So, let's see what the upcoming day's ideal applications of Environmental Computation look like.

We here raise several presumed but attainable applied scenes in Environmental Computation to help us understand how rationality could be detailedly specified by fathoming the relationship between the computations and environments in critical reasoning and mattering:

SCENARIO A - Visiting a complex of office buildings and business zones:

After a job seeker ends the interview and goes back to the lobby of the office building in which the hiring company is located in the complex scoped as a business zone (an environment in terms of Environmental Computation) possesses the capacity for Environmental Computation (EC), which consists of the High-Performance Computational Units (EC Nodes or Entities), the local Low-Latency Networks, and the App in EC installed in the device (Smartphone) of the interviewee, based on the configuration or set, whose smartphone may actively initial the EC App to prompt the interface listing a bunch of browseable items as shortcuts to guide she/he (a user of EC App) gaining the most desired services according to the purposed requirements elicited by the features of the location and the definition of the roles (stakeholders analysis) as the people among the environments.

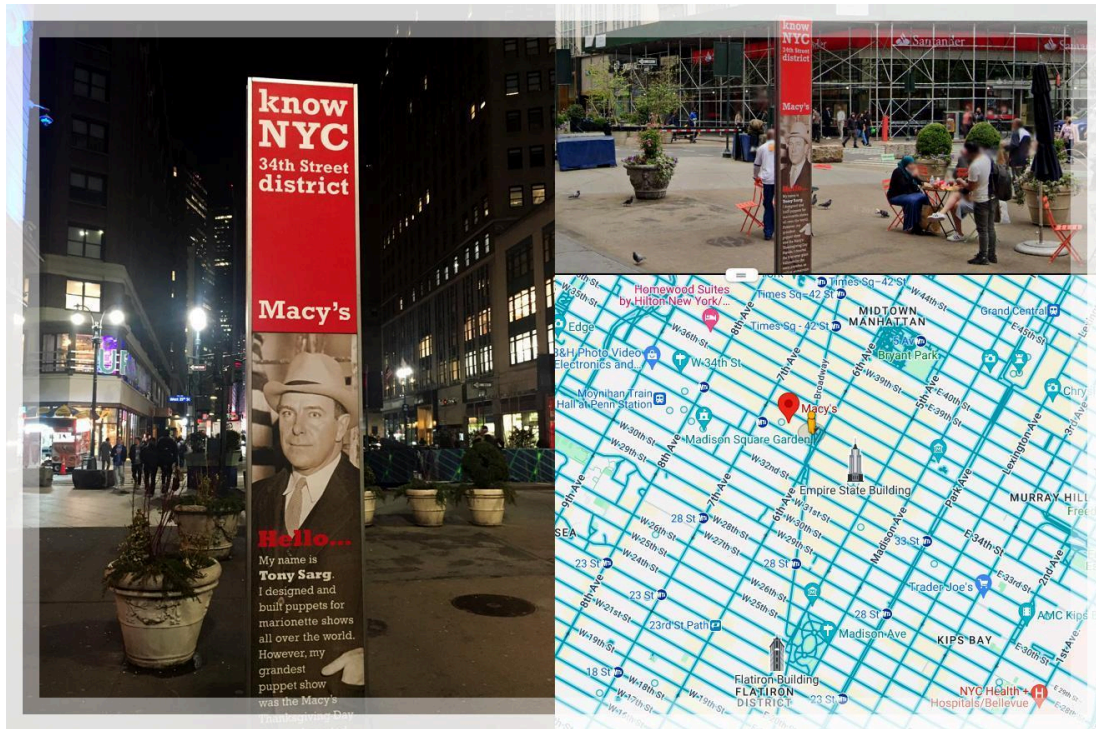
Essentially, each environment has its pivotal feature to fulfill the requirements of the people

among it such as a resort hotel purposely serves travelers, a business complex as the above exemplified generally provides the office personnel convenience, and a university campus typically satisfies the students and faculty's needs, whatever the very environment integrates the power of the Environmental Computation technology and the applications or not, the respective feature is always coherent to the discrepant environment, which would be never changed until the exact environment completely has been rooted and thence becomes a different place physically. Yet, one environment may still at least own its standard or so-called baseline to apply the Environmental Computation championship like a security system with a matrix of sensors such as CCTV cameras powered by facial recognition with Artificial Intelligence argument, which indicates that, despite an environment such as a hospital chiefly provides the medical treatments, which shall keep its Environmental Computation capacity guaranteed availability of a series of fundamental functionalities.

Here is a very interesting thought inspired by tangible stuff erected on the pavement in the City of New York, which is a historical commemorative lightbox sign, it introduces the fun stories with the relevant stakeholders that happened in the old days at the same spot nowadays. So, let's think of what if this stuff equips the capacity of Environmental Computation, what some of the further advanced abilities in applicational consideration may benefit the people loitering the area (the very environment) or passing by the spot?

SCENARIO B - Loitering the environment and passing by the lightbox sign spot:

_ See the next page _



On their first time in New York City, travelers loiter in the big area of Times Square near Macy's flagship store and pass by this lightbox sign intended to tell stories of the old days at the very locational spot. In such kind of circumstances, if the sign has been embedded in the capacity of Environmental Computation (EC), the lightbox sign shall have the ability to differentiate between, tourists, who are just quickly walking by or someone who stops in front of, by the further real-time analysis, if the sign realizes the situation via the sensor configured that the tourist is exactly staring the storyboard, reading the content with a potential interest to explore more background details behind the historical story, it, as one of the units deployed by this strategically mechanistic area powered the specific EC technology, should be on-time to waken the EC App on the smart device typically such as cellphones, prompting the tourist as an end user there's the availability to let her/him fetch furthermore interesting context to explore. We believe, naturally, a bunch of valuable information would be able to be delivered to the end users as the people immersed in the environment, the informative context could be the knowledge in cultural exploration, promotions in tourism events, elicitation in consumers purchasing willingness, etc.

According to this above scenario hypothesis, it has been proved that Environmental Computation does have a strong feasibility and a bright future to tremendously benefit people's daily lifestyles, requirements fulfillment, and particularly the business world with the technological stacks development in next-generation Internet and properly applied Advanced Artificial Intelligence. There's another envisage that may tell more about how the applications can perform profoundly in a critical situation. Let's see the following example.

SCENARIO C - Saving the life of a pedestrian passing by the unit of Environmental Computation (EC) in a critical situation of heart attack:

Assuming midnight, an elderly person walking on the sidewalk and passing by the EC unit deployed on the street suddenly gets a heart attack, lying down helplessly. Still, no one notices this emergency, and seemingly the tragedy inevitably happens. At this critical moment, the technology of Environmental Computation shall instantly take over to handle this life-saving crisis, through a kind of functionality of the so-called “Street Sentinel” embedded in the EC unit (a specific endpoint purposely disposed on the street corners), the sensor with mechanism shall understand the pedestrian badly needs an emergent CPR support, the system of this EC unit will soon manage to dispatch the life-saving requests to the surrounding, trying to reach out to someone who might be a doctor or professionals owning the legitimate aptitude in medical emergency treatment to save the person under the moribund status because of a sudden heart attack. At best, fortunately, a qualified cardiologist living nearby receives the notification and catches up to arrive at the location, implementing the cardiac resuscitation and finally saving the life.

So, despite supposing, the case with the above-raised examples logically and tenably reveals, in the specific circumstance, an application of Environmental Computation unquestionably owns a strong ability to handle a series of complicated scenarios, most significantly, reversing the tide of quandaries and saving the originally unavoidable plight. In practice, applying applicational Environmental Computation shall be feasible based on the present technological stacks in Computer Science, Electrical Engineering, City Science, Socio-Technical Systems Engineering, and especially Human-Computer Interaction. The reason why we still lack seeing EC applications in our surroundings nowadays is not because of technical or financial deficiencies but people’s mindsets. The stereotypical and prejudiced thoughts matter a lot yet the world constantly changes, in the vast majority of situations when encountering some potentially game-changing technologies or applications, most people naturally overlook the possibilities, benefits, profitabilities, development, and evolvement of human beings’ increasingly demanding requirements every day.

“500 dollars fully subsidized with a plan!? I said that is the most expensive phone in the world and it doesn’t appeal to business customers because it doesn’t have a keyboard...” said Steve Ballmer, the former CEO of Microsoft when the brand new iPhone came to this world. 10 years later, he said regretfully, “I wish I could be back to those days, well re-making smartphones.” Also, the once-dominant giant Nokia evaluated if the iPhone was going to challenge the market through a series of rigorous experiments, they concluded that the competitor iPhone would eventually fail since it would be highly possibly damaged if the phone slipped from the height of the pocket of the pants. Currently, we all see the unprecedented success of Apple with the iPhone production lines and fantabulous iOS platform, the estimation and judgment from these prestigious entrepreneurs evidenced the proven shortsightedness and satire. Sometimes, the moment of opportunities and changes remain vague and hard to recognize by the stakeholders, markets/industries, professionals, or even superbly qualified researchers or scientists, which means, even though a great amount of newly evolved and emerged requirements demanded by modern society, and most crucially, although

the readiness in engineering stacks for materializing a brand new form of the applied technology has been fully prepared, the realm of business and technology may still be devoid of pioneers and encouragement. Perhaps, the mattering and reasoning behind the present quandary or dilemma of the development of Environmental Computation or related fields such as City Sciences, Human-Computer Interaction, Socio-Technical Systems Engineering, and so on are legitimately complex. Still, our competence to alter the situation and facilitate bringing cutting-edge applicational technology, like the very one here introduced “Environmental Computation”, to deliver the public’s daily satisfaction with modern technological finesse is ensured as long as we deeply apprehend the importance of research in Human-Centered Methodology and innovatively leverage the currently existing and developing technologies such as Electrical Engineering, Computer Science/Engineering, Software Engineering, Data Science, Machine Learning, Applied Mathematics, and City Science and even with Civil Engineering.

As we repeatedly emphasized above, the notion of Environmental Computation is not so hardly comprehended, which though seemingly looks like one transdisciplinary suture that may be contingently integrated with “Farfetched” ideas. On the other hand, the feasibility of this “Utterly Novel” technology with applications has already secured the prerequisite in actualization based on a vast range of contemporary established and developing technological stacks and even so-called high-end applied tactics in the Digital Era e.g. Small-Area Covered Low-Latency Networks in 6/7G, Low-Power Consumption Sustainability, Facial Recognition, User Gesture Capture/Understanding, High-Performance Local/Cloud Computations, etc. With full confidence, we can say today the possibility of instantiation in diversified applications of Environmental Computation has been affirmatively presented in front of us, the benefit is indeed and the future is now. Unlike some habitually authored academic research papers that were inclined to indicate the “Urgency” of the raised conceptual idea for technological innovation, we intend to thoroughly figure out what exactly the “Necessity” of Environmental Computation is, and thence, the feasibility including the consideration of prerequisites should be smoothly elicited to ultimately lead to the conclusion as a set of guidance sharing and facilitating the after experiments for our colleagues, researchers, and scientists, hoping we together embrace a bright future.

NECESSITY

3.01 - Questioning, Debating, and Understanding

“Is this paper about an idea, conception, envisage, even new technology, or just something else like these?”, This was the questioning response after sharing the thesis by our corresponding author Kong to a talented newly admitted master’s student shortly studying at NYU in Fall 2024.

“Why should I be willing to use applications by taking advantage of this kind of new tech instead of nowadays well-established technologies on the existing Internet, could it be more powerful or convenient for me?”, which was the concern raised by a superbly experienced industrial practitioner who’s in the field of Software Development and DevOps Engineering for many years.

These were reasonable responses and reflections of the people from the relevant professional area in research and practice, they have of course concerns regarding technological actualization, more profoundly, and applicational necessity via the perceptions of their expertise. Yet, the persons from business analytics and operation research hold comparatively positive opinions, they generally think this object, the technology of Environmental Computation and its applicational scenarios, ought to be a tenably foreseeable pattern in very near days as one of the applied science stuff when our society get readiness in evolvement with the conditions of modern technologies perfection and business advancement. Nonetheless, mundane people without a fulfilled background in Science, Technologies, Engineering, and Mathematics (STEM) spontaneously confused and presented an insufficient understanding of the core concept of Environmental Computation, particularly the prospect of its applicational development. Interestingly, professionals from the fields of Human-Computer Interaction and Human-Centered Design in User Experience research and implementation possess a huge amount of curiosity and admiration, and they expect a future of potentially broad application possibilities in Environmental Computation. The following selected samples via questionnaires and statistics may tell more as a glimpse.

> Insert questionnaires and statistics here:

The survey expressed various viewpoints from different people holding discrepant academic or industrial backgrounds, which evidenced the phenomenon that questions the necessity of experiment, implementation, and investment for Environmental Computation remains controversial. Essentially, these reservations and somewhat negative opinions are naturally from people’s superficial cognition or insufficient comprehension of today’s high-velocity momentum in human society, which is changing and challenging. Throughout human beings’ modern and contemporary technological development, we witnessed that considerable once-brand-new stuff initially believed unnecessary eventually got accepted, admired, and become indispensable and irreplaceable today. Like the first automobile or railway train, in the beginning, when they were first introduced to people’s lives, the questioning of necessity dominated for a long while. We also see, that not long before, the release and following user experience and feedback of Apple’s Vision Pro brought ubiquitous disputations, some believe it was never before revolutionary, yet some think it was kind of redundant; apparently, this is a representative case intensively bonding to public’s opinions in “Necessity”. Having to admit, any of the newly introduced stuff in technological innovation at whatever historical moment inevitably endures its query of necessity; yet, with the days moving forward, people’s questioning became diminished.

In general, we may comprehend it’s because just time’s changing and people’s mindsets evolved, but in-depth, it’s the factor that the originally underestimated or ignored requirement(s) got

increasingly fulfilled and finally satisfied by the newly developed tech stuff, which essentially is the technological solution to each possible aspect in wide human society's activities and needs. Interestingly, we investigated or simply saw that none of the brand-new technological things obey the same time point arrival pattern in history or present days, some came to this world early and some came later; the above-raised example Apple Vision Pro shall be the early-coming one, and online-buying website such as Amazon may be the later one, we so understand no matter the arrival was leading or lagging, always, the essentiality of mattering, reasoning, and driving factors between technology development and stakeholder requirements never changes. Thus, at this watershed to a clearer explanation, quantifying the term "Necessity" by a series of mathematical models shall be an utter solution to fully fathom why the technology "Environmental Computation" and other once or present cutting-edge technological stuff are truly needed, which means the term "Necessity" is worth diving into, debating, and to thence confirm via repeated verifications and validations in a responsibly academic manner. So, the first equation as the following may tell us some:

$$N = \frac{1}{\log_T R}$$

From the above identical equation, we see, in a mathematical way, that the numeric value of the term "Necessity" can be quantified by a fraction's computation which has a logarithm denominator. First things first, regardless of the computation of the fraction, in the denominator, where the term T as a base means the numeric value of the conceptual term of "Technology" or so-regarded "Applied Tech Stuff" with its Technological Capacity, meanwhile, where the term R symbolizes the quantified value of the term of "Requirement" or so-meant "Stakeholders Requirement", which represents the exact meaning of some specific demand from human society in the real world.

For a clearer understanding of the coupling linkage relationship between the terms "Technology" and "Requirement", we may now numerically assign the values to both of them as the following equation:

$$N = \frac{1}{\log_3 9}$$

On the above, assuming that, there's an existing "Technology" that owns its quantitatively technological capacity's value 3, in addition, the term "Requirement" has its value 9; so that, the denominator's numerical value ought to be 2 by the logarithm computation, thence, we get the counted value of the term "Necessity" which is $\frac{1}{2}$ also 0.5 in decimal number.

So, at this moment, where, according to the equation or so-meant the computation of

quantifying the “Necessity”, raises a concept with the question: Why do we define a fraction and set a numerator as the value 1 and a denominator as the logarithm term to quantitatively figure out the value of the term “Necessity”? We believe there are a bunch of factors, one of the reasons for setting the very fraction holding a numerator assigned a constant value of 1 is to define a constrain, letting us fathom, attain, and thence ensure a critical concept in the classic engineering’s measurement methodologies that the quantitatively calculated value of the term “Necessity” cannot be a freely-set one, which ought to reflect a profoundly crucial notion to indicate how big or small number of a “Necessity” value could be reasonable, acceptable, affirmative, and decisive, or ideally balanced in perfection that follows the principles in engineering methods of estimation, evaluation, and applicational feasibility. We may reason more about why we set such a constant value of 1 from the below:

$$N = \frac{1}{\log_{10} 10}$$

Let’s say, that in the logarithm computation of the above equation, we assigned a numeric value of 10 to both terms of base and anti-logarithm so that the quantified term “Necessity” value is 1 exactly. Under such a conditional explanation, we say this “Necessity”, in a quantitatively and qualitatively analytic way, is a superbly balanced value in perfection, which in other words means, given the value of 1, we get a fact that this kind of “Technology” fully fulfilled this type of “Requirement”, furthermore implicitly indicates the existing or potentially applicable “Technology” of course is worthy to introduce, invest, and develop because of the existence of the stakeholders’ “Requirement”. With that, we simultaneously apprehend given the quantified evaluation of “Necessity” is a value of 0.978 that suggests this “Technology” is *quite* tenable in the necessity of existence, despite we yet don’t know the quantified value of “Technology” and “Requirement”, still, via a reversed reasoning way, we realize the two values of “Technology” and “Requirement” in quantitative are pretty close, which explicitly means this particularly needed “Technology” basically do or is satisfying this specifically scoped “Requirement”.

Based on the former analysis, already we comprehended, maybe in a simpler way regardless of the logarithm computation, that the ratio value quantified by measuring the values of “Technology” and “Requirement” decisively matters in the conclusion of how a quantitative and qualitative “Necessity” of a specified “Technology” could be. And, we may see this from the below-exemplified equations by respectively assigning the values of the terms “*T*” and “*R*” (Technology” and “Requirement):

(a)

$$N = \frac{1}{\log_2 16} = 0.25$$